Interactive training for functional analysis and interpretation of disease data using computational modelling tools

Lead Instructor - Dr. Anna Niarakis – UT3 & INRIA, France

Full Professor, HDR
PI CoSysBio,
CBI-CMD, University of Toulouse
& Lifeware, INRIA, Saclay

https://cbi-toulouse.fr/fr/equipe-niarakis

anna.niaraki@utoulouse.fr
Anna Niarakis

Research Focus

- Computational Systems Biology for complex human disease
- Disease maps construction and analysis
- Large scale Boolean modeling of signaling networks
- Curation and Annotation of Logical Models in biology
- Hybrid models
- Digital twins in healthcare

Université de Toulouse

Lifeware Inria informatics mathematics
Computational Systems Biology and Optimization

CoSysBio

cbi CENTRE DE BIOLOGIE INTÉGRATIVE

Building Immune Digital Twins

systems medicine disease maps

Lead Instructor - Dr. Ben Hall– Medical Physics and Biomedical Engineering, University College London

Current:
UCL, Co-chair Computational Cancer Collaboratorium

Previous:
Royal Society Research Fellow/ MRC Investigator University of Oxford, Microsoft Research

B.hall@ucl.ac.uk
Benjamin A Hall

Research Focus

- Computational modelling of the early stages of carcinogenesis
- Executable modelling of cancer networks (ion channels, signalling, epigenetics)
- Spatial models of clone competition and expansion
- Molecular modelling of protein mutations in cancer

THE ROYAL SOCIETY

UCL

CANCER RESEARCH UK

https://www.hall-lab.com

Welcome!

- Wellcome Genome Campus course on discrete modelling in biology, focusing on network biology & human disease -4 edition!
- Second time in person!

- **Our goal:**
  - Provide a grounding in diverse set of tools from across the field of network biology and discrete dynamic modelling
  - Back tools and approaches with a series of lectures and seminars showcasing success stories from leaders in the field

Systems biology and biomedicine

- Multiple challenges and opportunities
  - Expansive databases derived from patient data
  - Detailed interaction maps available
  - CRISPR makes genetic manipulation easier and more routine

- Underlying mechanisms remain elusive
  - Necessary for translation and understanding
  - Identify fundamental organisational principles to understand wider systems

Executable modelling

- Approach building on logical modelling and computer science in formal verification
- Driven by high level abstractions focused on functional activity

- Simplified relationships
- Model checking

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Structure

Keynote Seminars	Flagship successes in the field
Lectures	Introduction and background theory
Practicals and demos	Hands on learning/ introducing tools
Projects	Application of tools and concepts from the course

Organisation

- Anna & Ben
  - Leads- always available
- Trainers
  - Lead lectures, practicals
- Wellcome Connecting Science Team
  - Technical support
- Keynotes
  - Seminars

Keynote speakers

Sr Sarah Teichmann

Professor Sarah Teichmann, FMedSci FRS
Laboratory: Cambridge Stem Cell Institute, Jeffrey Cheah Biomedical Centre

Dr Henning Hermjakob

Head of Molecular Systems services at EMBL-EBI, which provide worldwide reference data resources in interactomics (IntAct), pathways (Reactome), and systems biology models (BioModels).

Dr Jasmin Fischer

Professor of Computational Biology at UCL Cancer Institute

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Trainer - Dr. David Shorthouse – University College London

Lecturer in Pharmaceutics
UCL School of Pharmacy

Co-founder - Implexis Ltd

d.shorthouse@ucl.ac.uk
David Shorthouse

Research Interests

- Drug response in oncology
- Drug repurposing (With a drug in clinical trials)
- Integrating computation with automation and robotics
- Deep learning for drugs

https://profiles.ucl.ac.uk/82451-david-shorthouse

www.implexis.ai

Trainer - Dr. Sylvain Soliman INRIA-Saclay

Computer Science Researcher (CR)
Lifeware Team
INRIA, Saclay

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Research Focus

- Computational Systems Biology
- Formal methods and Petri nets
- Boolean and continuous models
- Constraint Programming

Lifeware
Computational Systems Biology and Optimization

Inria informatics mathematics

Trainer - Dr. Tomas Helikar – University of Nebraska - Lincoln

Associate Professor
Department of Biochemistry

Research Focus

- Multi-scale modeling of the human immune system
- Software development (e.g., Cell Collective, ccNetViz)
- Hands-on modeling as a method (instead of memorization) to learn about biology in life sciences courses

http://helikarlab.org

https://cellcolective.org

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@helikarlab @tomashelikar

Trainer - Dr. Pedro Monteiro – IST – Universidade de Lisboa

Associate Professor
Department of
Computer Science and
Engineering

Research Focus

- Logical modeling of regulatory networks
- Boolean functions & Formal model revision
- Integration of regulatory / metabolic models
- Software development (e.g., EpiLog, GINsim)

http://pedromonteiro.org

Pedro.Tiago.Monteiro@tecnico.ulisboa.pt

@ptgmonteiro

http://ginsim.org

http://epilog-tool.org

http://colomoto.org

Trainer - Dr. Vincent Noël – Institut Curie, Paris

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Institut Curie

Computational Systems Biology of Cancer

CBIO - Mines ParisTech
INSERM U900
PSL Research University

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Vincent Noël
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Research Focus

- Modeling of Biological Systems
- Boolean modeling
- Multiscale modeling
- MaBoSS, PhysiBoSS developper
- High Performance Computing

institut Curie

Inserm
La science pour la santé
From science to health

MINES PARIS

PSL
RESEARCH UNIVERSITY PARIS

https://curie.fr/personne/vincent-noel, https://vincent-noel.fr

Trainer - Dr. Rahuman Sheriff – EMBL-EBI

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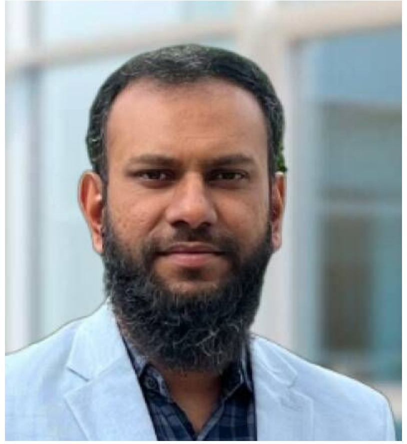

Senior Project Leader, BioModels
European Molecular Biology Laboratory
European Bioinformatics Institute (EMBL-EBI)

Affiliate Associate Professor
University of Florida

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Rahuman S. Malik Sheriff
@rsmsheriff

Research Focus

- Modeling of Biological Systems
- BioModels Repository
- Reproducibility, SBML and Standards
- Hybrid modelling
- Rare Disease Digital Twins

EMBL-EBI BioModels CHAN ZUCKERBERG INITIATIVE UF UNIVERSITY of FLORIDA

https://directory.ufhealth.org/malik-sheriff-rahuman-sheriff

Trainer – Dr. Arnau Montagud – Institute for Integrative Systems Biology (I2SysBio), CSIC-UV

Systems Biology Modelling lab
I2SysBio
Spanish National Research Council (CSIC)

Visiting Researcher at Barcelona Supercomputing Center (BSC)

Research Focus

- Digital twins in healthcare
- Boolean modelling
- Multiscale modelling
- High Performance Computing (HPC)

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CONSEJO SUPERIOR DE INVESTIGACIONES CIENTÍFICAS

https://arnaumontagud.netlify.app

Trainer – Dr. Martina Summer-Kutmon - Maastricht University, NL

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Assistant professor
Maastricht Centre for Systems Biology & Bioinformatics (MaCSBio)

WikiPathways architect
ISCB NetBio COSI representative

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Martina Summer-Kutmon
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Research Focus

- Network biology
- WikiPathways
- Pathway models
- Multi-omics data integration
- Digital twins in healthcare
- Data visualization
- Software development: PathVisio, CyTargetLinker, WikiPathways

Maastricht University

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NetBio

WikiPathways
Pathways for the People